

ITM SKILLS UNIVERSITY

School of Future Tech

JAVA PROGRAMMING PROJECT REPORT

Warranty Ka Kya Scene Hai?

Product Warranty & Service Claim Management System

Course	Java Programming
Semester	III
Program	B.Tech CSE 2025–29
Project Type	Case Study 82
Technology	Java, Swing
Application Type	Desktop GUI Application

Prepared by: Sanika Kangane

1. Introduction

Warranty Ka Kya Scene Hai? is a Java-based Product Warranty & Service Claim Management System developed to help an electronics retailer manage products, warranty information, customer service claims and repair appointments through a simple desktop graphical user interface. The application provides a centralized way to register products, track warranty periods, file claims, schedule repairs, search records and identify expired warranties.

The project demonstrates important Java programming concepts including object-oriented programming, abstraction, method overriding, collections, CRUD operations, searching, sorting, exception handling, validation and Java Swing GUI development.

2. Problem Statement

An electronics retailer needs a system to track product warranty periods, manage customer service claims and schedule repair appointments across different product categories. The system should also flag products with expired warranties.

3. Objectives

- Maintain product and warranty information.
- Register products with customer and purchase details.
- Manage customer service claims.
- Schedule repair appointments.
- Identify and flag expired warranty claims.
- Search product and claim records.
- Sort records by relevant dates such as warranty expiry and claim date.
- Provide CRUD operations for warranty and claim management.
- Provide a user-friendly warranty management GUI using Java Swing.
- Apply validation and exception handling to prevent invalid data.

4. Expected Modules

No.	Module	Purpose
1	Product Registration	Register product ID, product name, serial number, purchase date, customer and category.
2	Warranty Tracking	Calculate and track warranty duration according to product category.
3	Claim Filing	Create and manage customer service claims using a unique Claim ID.
4	Repair Scheduling	Schedule repair appointments for valid service claims.
5	Expiry Alerts	Identify claims/products whose warranty has expired.

6	Claim Search	Search product and claim records.
7	Service Reports	Display registered products, claims and service-related information.

5. System Design and Main Classes

The system is organized using separate Java classes so that each class has a clear responsibility. The main domain classes represent products, warranty rules, claims and service-center operations, while the GUI class handles user interaction.

Class	Responsibility
Product	Abstract/base product class containing common product information and warranty behavior.
Electronics	Product category that provides category-specific warranty behavior.
Appliance	Product category that provides category-specific warranty behavior.
Gadget	Product category that provides category-specific warranty behavior.
Warranty	Stores warranty-related information and expiry details.
Claim	Represents a customer service claim and its related details.
ClaimStatus	Represents the state of a claim such as pending, approved or completed.
ServiceCenter	Manages products, warranties, claims, searching, sorting and service operations.
ExpiredWarrantyException	Custom exception used when a claim is attempted against an expired warranty.
WarrantyManagementGUI	Java Swing interface used to interact with the complete system.
Main	Entry point used to start the application.

6. Java Concepts Implemented

Java Concept	Application in Project
Classes & Objects	Product, Warranty, Claim and ServiceCenter objects model the real-world system.
Constructors	Initialize product, warranty and claim objects with required data.
Abstract Class	Product acts as the base class for different product categories.

Method Overriding	Category-specific classes override warrantyPeriod() to provide different warranty rules.
ArrayList	Stores collections of products and claims.
LinkedList	Maintains claim history in sequence.
HashMap	Maps Product ID to warranty/product-related information for quick lookup.
TreeMap	Maintains claims in sorted key order, useful for date-based organization.
CRUD	Create, read, update and delete operations are used for record management.
Searching	Records can be located using product or claim information.
Sorting	Records can be sorted by warranty expiry or claim date.
Swing	Java Swing components create the desktop warranty management GUI.
Exception Handling	Handles expired warranties, invalid claims and invalid input.
Validation	Validates serial number, purchase date and claim-related information.

7. Product Registration

The Product & Warranty section is used to register a product. The user enters a Product ID, Product Name, Serial Number, Purchase Date, Customer Name and Product Category. The category selection determines the warranty rule applied to the product.

Typical registration information includes:

- Product ID – unique identifier for the product.
- Product Name – name of the electronic product.
- Serial Number – unique serial number of the physical product.
- Purchase Date – date on which the product was purchased.
- Customer Name – customer associated with the product.
- Category – Electronics, Appliance or Gadget.

8. Warranty Management

Warranty management uses abstraction and method overriding. The Product base class defines the common structure, while Electronics, Appliance and Gadget can provide category-specific warranty periods.

For example, if a product is registered on 2026-01-07, its warranty expiry is calculated from the category-specific warranty period. A repair appointment is scheduled separately and normally occurs after a claim has been filed.

9. Claim Management

A service claim represents a customer's request for repair or service. Each claim is assigned a unique Claim ID so that it can be tracked independently.

- Claim ID identifies a specific service request.
- Product ID connects the claim to the registered product.
- Claim details describe the reported problem.
- Claim status indicates the current stage of the request.
- The system checks warranty validity before processing an eligible claim.

10. Repair Scheduling

After a valid claim is created, a repair appointment can be scheduled through the service-center workflow. The repair date is independent of the original registration date. For example, a product registered on 2026-01-07 could have a claim filed later and a repair appointment scheduled after the claim date.

11. Graphical User Interface

The application uses Java Swing to provide a desktop interface. The main interface is designed around three functional areas: Product & Warranty, Claims & Repair, and Search & Reports.

- Product & Warranty – product registration and warranty information.
- Claims & Repair – service claim creation and repair-related operations.
- Search & Reports – searching, sorting and viewing system output.
- Clear buttons allow the user to reset input fields.
- Register Product and other action buttons use visible bordered styling for clear interaction.
- The main heading 'Warranty Ka Kya Scene Hai?' gives the interface a simple, modern and memorable identity.

12. Data Structures Used

Different collection types are used according to the purpose of each operation:

- ArrayList: suitable for maintaining an expandable list of products and claims.
- LinkedList: suitable for sequential claim history where insertion and history traversal are important.
- HashMap: provides fast lookup using a key such as Product ID.
- TreeMap: maintains entries in sorted key order and supports organized claim/date information.

13. Exception Handling and Validation

Exception handling improves reliability by preventing invalid operations from terminating the application. The custom ExpiredWarrantyException is used to represent an expired warranty condition.

- Invalid or incomplete product information is rejected.
- Invalid date formats are handled using date parsing and validation.
- Expired warranty conditions are detected before eligible claim processing.
- Invalid claim information is handled through validation and exceptions.
- User-friendly messages are displayed through the GUI.

14. CRUD Operations

Operation	Use in System
Create	Register a product or create a service claim.
Read	Display registered products, warranty details and claim records.
Update	Modify relevant claim or service information.
Delete	Remove records when required by the service workflow.

15. System Workflow

1. Start the Java application.
2. Open the Product & Warranty section.
3. Enter product and customer details.
4. Select the product category.
5. Register the product and calculate applicable warranty information.
6. File a service claim when a product develops a problem.
7. Validate the warranty and claim details.
8. Schedule a repair appointment for a valid claim.
9. Search or sort records when required.
10. Display service information and reports.

16. Sample Scenario

Consider a customer named Aarav who purchases a laptop. The product is registered in the system with Product ID P101 and Serial Number SN-LAP-101. The product category is Electronics and the purchase date is 2026-01-07.

Product ID	P101
Product Name	Laptop
Serial Number	SN-LAP-101
Purchase Date	2026-01-07
Customer	Aarav
Category	Electronics
Example Claim ID	C001
Issue	Laptop screen is not working
Example Repair Date	2026-01-10

17. Testing

Test Case	Action	Expected Result
Valid product registration	Enter complete valid product details	Product is registered successfully
Missing field validation	Leave a required field empty	System displays validation message
Invalid date	Enter an invalid date format	System rejects the date
Valid claim	Create a claim for a valid product	Claim is created successfully
Expired warranty	Attempt a claim after warranty expiry	ExpiredWarrantyException / warning is triggered
Search	Search by Product ID or Claim ID	Matching records are displayed
Clear	Click Clear after entering data	Input fields are reset
Sorting	Sort records by relevant date	Records appear in sorted order

18. Advantages of the System

- Centralizes product, warranty and service claim information.
- Reduces manual tracking of warranty periods.
- Makes claim records easier to search and manage.
- Supports category-specific warranty rules.
- Improves reliability through validation and exception handling.
- Provides a simple GUI suitable for non-technical users.
- Demonstrates multiple core Java concepts in one practical application.

19. Future Enhancements

- Connect the application to a database such as MySQL or PostgreSQL.
- Add customer login and service-center staff login.
- Send automatic warranty expiry notifications.
- Add email or SMS notifications for repair appointments.
- Generate downloadable service reports.
- Add dashboard cards for active claims, completed repairs and expired warranties.
- Add role-based access control for customers and service-center staff.

20. Conclusion

Warranty Ka Kya Scene Hai? successfully demonstrates how Java can be used to build a practical warranty and service claim management application. The project combines object-oriented programming with Java collections, abstraction, method overriding, CRUD operations, searching, sorting, validation, exception handling and Swing GUI development. The resulting system provides a structured workflow for product registration, warranty tracking, claim management and repair scheduling.

Appendix A – Project File Structure

```
src/
├── Product.java
├── Electronics.java
├── Appliance.java
├── Gadget.java
├── Warranty.java
├── Claim.java
├── ClaimStatus.java
├── ServiceCenter.java
├── ExpiredWarrantyException.java
├── WarrantyManagementGUI.java
└── Main.java
```

End of Report